Development of a barrier-based control strategy for the safe navigation of mobile robots.

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***Abstract*—This work presents a control strategy to ensure the safe navigation of a differential mobile robot during target-tracking tasks. The proposed cascade control architecture integrates a position controller; a safety controller based on Control Barrier Functions (CBF) and Control Lyapunov Functions (CLF) formulated as a quadratic programming problem (QP); and a current controller implemented on the microcontroller STM32 F446RE. In addition, two Kalman filters are employed for state estimation: one designed for estimating the motor currents and another for fusing the information of pose from odometry sensors and the robot's dynamic model. The proposed strategy was validated through simulations and experimental tests on a differential robot. The experimental results demonstrate that the system follows the target, satisfying safety constraints imposed by the control barrier function throughout the optimization process, verifying the Lyapunov and barrier conditions compliance that constitute the formal certificates of system stability and security.**

***Index Terms*—Control Barrier Function, Control Lyapunov Function, differential robot, safety control, Kalman filter**

I. Introduction

Robots operating near people must balance tracking performance with human safety [1], [2]. In differential-drive robots, human tracking is challenging because the reference varies with human behavior and must be estimated from sensory information [3]. Although previous approaches address tracking stability and obstacle avoidance [4]–[6], accurate tracking does not necessarily ensure safe separation from the person [7].

Control Lyapunov functions describe convergence objectives, whereas control barrier functions enforce safety through forward invariance. Their integration into a quadratic program (CLF-CBF-QP) balances nominal control objectives and safety constraints [8]. Although this approach has been applied to multirobot systems, kinematic control, and human-robot interaction [9]–[13], implementations for differential-drive robots formulate safety using kinematic models [14], [15], limiting the explicit consideration of robot dynamics and actuator constraints.

This work proposes a hierarchical architecture comprising a position controller, a dynamic CBF-based safety filter, and a wheel-torque controller. By defining the barrier function in terms of linear velocity, its derivative incorporates the corresponding acceleration, enabling the QP to account for robot dynamics and explicitly modeled actuator limits. The architecture is experimentally tested using a virtual moving target to evaluate tracking and safety-related performance.

Section II presents the safety filter and its optimization problem; Section III describes the system implementation; Sections IV and V presents and discusses the simulation and experimental results; and Section VI concludes the paper and outlines future work.

II. Safety Control

Considering the following control-affine dynamical system:

$$\begin{aligned} \dot{x} &= f(x) + g(x)u, \\ y &= h(x), \end{aligned} \tag{1}$$

where x denotes the state vector, u is the control input, and y is the system output.

A. Control Lyapunov Function

A defined positive and continuously differentiable function allows establishing the stability and convergence of a system to an equilibrium point. Considering the dynamic system proposed in the equations 1, this will be asymptotically stable at an equilibrium point $x^* = 0$, if there exists a function $V(\mathbf{x})$, defined positive, such that $V(\mathbf{x}) \geq 0$ for all $\mathbf{x}$. Due to the complexity of function formulation that complies with these kinds of properties, exponential convergence is imposed through the following constraint about the temporal evolution of $V(\mathbf{x})$ [16], [17]:

$$\dot{V}(x) = \nabla V(x)\dot{x} \leq -\lambda V(x). \tag{2}$$

When substituting the system dynamic in the previous expression, the stability constraint is obtained:

$$L_f V(x) + L_g V(x)u \leq -\lambda V(x), \tag{3}$$

where $L_f V(x)$ and $L_g V(x)$ denote the Lie derivatives of $V(x)$ the vector fields $f(x)$ and $g(x)$, respectively, as defined by the affine system in (1).

B. Control Barrier Function

For their part, the control barrier functions (CBF) are used to guarantee that the system state remains within a predefined safe set C. This set corresponds to a continuously differentiable one, whose borders ∂C define the safety limits and must comply with the following conditions:

$$\mathcal{C} := \{x \in \mathcal{X} : B(x) \geq 0\},\ \partial\mathcal{C} := \{x \in \mathcal{X} : B(x) = 0\}. \quad (4)$$

A sufficient condition for the forward invariance of the safe set is given by:

$$\dot{B}(x) \geq -\gamma B(x), \quad (5)$$

where $\gamma > 0$ determines the rate at which the system is driven toward the interior of the safe set. For the control-affine system, the barrier function dynamics are expressed as

$$\dot{B}(x) = L_f B(x) + L_g B(x) u, \quad (6)$$

replacing (6) with (5) yields the inequality constraint that enforces forward invariance of the safe set and is incorporated into the quadratic problem.

C. Quadratic Programming Formulation

Taking into account the imposed constraints by the CLF and the CBF, the control action is calculated by solving a quadratic programming problem. To ensure viability, a slack variable is introduced, δ, exclusively in the CLF restriction, which allows relaxing the stabilization target when the safety and stabilization requirements become incompatible. In consequence, the optimization process gives priority to the compliance of the safety constraint imposed by the CBF and to the time that preserves the stabilization target when it is feasible. The optimization problem resulting is formulated following what is proposed in [17] as

$$\begin{aligned} \min_{u,\delta} \quad & w_u \|u\|_2^2 + w_\delta \|\delta\|_2^2 \\ \text{s.t.} \quad & \begin{cases} L_f V(x) + L_g V(x) u \leq -\lambda V(x) + \delta, \\ L_f B(x) + L_g B(x) u \geq -\gamma B(x). \end{cases} \end{aligned} \quad (7)$$

III. Implementation

Initially, the architecture of 1 is presented, which integrates a position controller, a safety controller, and a current controller in addition to a Kalman filter implemented in the microcontroller and the sensor fusion module for trajectory tracking.

A. System Model

Starting from the scheme shown in Fig.1, it is necessary to define the model of the system to be analyzed, which is a differential mobile robot with a non-uniform mass distribution, whose pose is defined by the coordinates $q = [x, y, \theta]$ with respect to point P_o, where x and y denote the Cartesian position expressed in meters (m), and θ represents the orientation of the robot, given in radians rad with respect to the x-axis. Furthermore, the linear velocity is expressed in meters per second (m/s), the angular velocity in radians per second (rad/s), and the control inputs, whose units are given in Newton-meters (Nm), correspond to the torque of each motor obtained through the dynamic model developed below.

The kinematic model is defined as follows:

$$\dot{q} = S(q)\eta, \quad (8)$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -d\sin(\theta+\gamma) \\ \sin(\theta) & d\cos(\theta+\gamma) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}, \quad (9)$$

where v and ω of the vector η are the linear and angular velocities of the system, respectively, and represents the deviation angle of the center of mass. The robot's dynamic model is derived in [18] using Rosenberg's formulation [19], which gives rise to the following simplified dynamic representation:

$$\bar{M}\dot{\eta} + \bar{V}_m \eta + \bar{F}_r \eta = \bar{B}\tau \quad (10)$$

where $\bar{M} \in \mathbb{R}^{2\times2}$ is a symmetric positive-definite inertia matrix, $\bar{V_m} \in \mathbb{R}^{2\times2}$ represents the Coriolis and centripetal effects, and $\bar{B} \in \mathbb{R}^{2\times2}$ is the input transformation matrix. Additionally, the matrix (F) is incorporated to model the dissipative forces acting on the velocity vector η.

Continuing with the DC motor model, a mesh analysis is carried out to formulate the following state-space representation of the DC motor [20]:

$$\begin{bmatrix} \dot{\omega} \\ \dot{i} \end{bmatrix} = \begin{bmatrix} -\frac{B}{J} & \frac{K_m}{J} \\ -\frac{K_m}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} \omega \\ i \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} v, \quad (11)$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \omega \\ i \end{bmatrix}. \quad (12)$$

B. Parameter identification

Once the models of the direct current motor and the differential robot have been established, the parametric identification of the plant is carried out, in which a constant torque is applied to each of the motors in order to analyze the behavior of the state variables and determine their influence on the system dynamics. Thanks to an optimization process, the parameters listed in Table I were obtained. It is worth mentioning that the data were obtained using the Jetson AGX Orin platform and the STM32 microcontroller; subsequently, they were stored in text files for processing and analysis.

TABLE I: Robot Parameters

Parameter	Units	Value
Mass	kg	3.25
Wheel radius	m	0.05
Wheel separation	m	0.35
Center of mass offset	m	0.08
Moment of inertia	kg·m^2	0.085
Torque constant	$\frac{\text{k·N·m}}{\text{A}}$	0.0285
Armature inductance	mH	50
Armature resistance	Ω	500
Moment of inertia	kg·m^2	0.0295
Viscous friction coefficient	$\frac{\text{N·m·s}}{\text{rad}}$	0.3

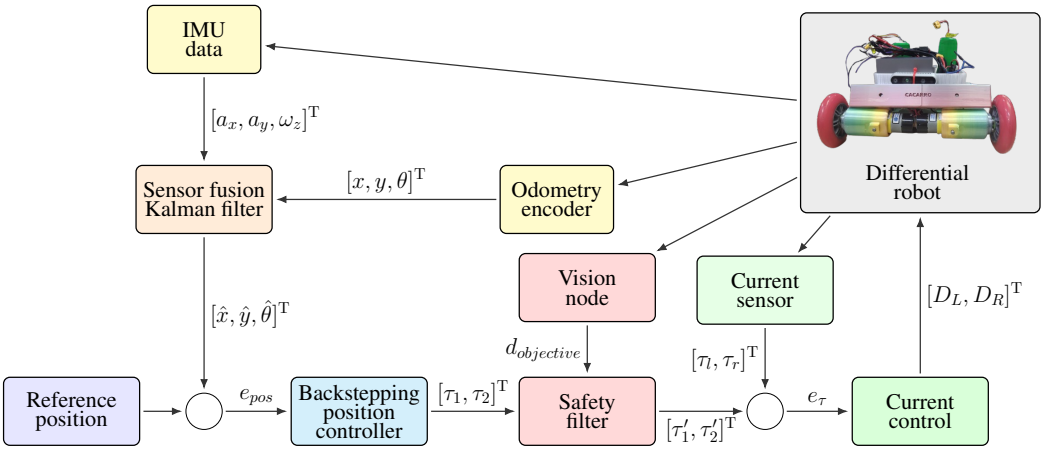


Fig. 1: Complete control architecture for a differential mobile robot.

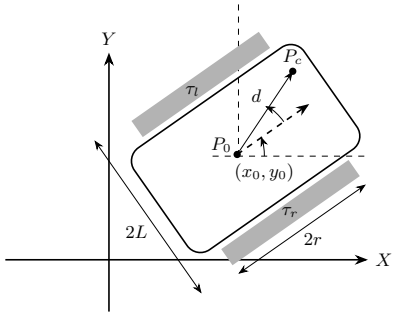


Fig. 2: Geometry of the differential-drive mobile robot.

C. Control Scheme

The proposed control architecture follows a cascade structure composed of a high-level controller and a low-level controller. The outer loop calculates torque references from the position controller and adjusts them using a safety controller based on barrier functions to ensure the simultaneous fulfillment of the tracking task and safety constraints. Meanwhile, the inner loop uses these references to regulate motor current, ensuring their tracking through a low-level controller. The estimation of the states required for control is obtained through a sensor fusion scheme, while the exchange of information between both levels is carried out via ROS2 and Micro-ROS.

1) Position Controller: The proposed position controller is a trajectory-tracking controller based on the dynamic model of the differential-drive robot. First, a computed-torque formulation is introduced to compensate for the robot dynamics, as given by (13). Afterwards, the tracking error is expressed in the robot reference frame, allowing the computation of the virtual linear and angular velocity commands required to drive the robot toward the desired trajectory, as defined in (14). Finally, the auxiliary control input is computed according to (15). Following the approach presented in [18], [21], the auxiliary control law is defined as follows:

$$\tau = \bar{B}^{-1}(q)\left[\bar{M}(q)\dot{\eta} + \bar{V}_m(q,\dot{q})\,\eta + \bar{F}_r(q,\dot{q})\,\eta\right], \qquad (13)$$

$$v_c = \begin{bmatrix} v_r\cos(e_3) - \omega_r d\cos(\gamma)\sin(e_3) + K_x e_1 \\ \omega_r + v_r\left(K_y e_2 + K_\theta \sin(e_3)\right) \end{bmatrix}, \qquad (14)$$

$$u = \dot{v}_c + K\left(v_c - \eta\right). \qquad (15)$$

The equation (13) includes an additional friction term to model dissipative effects in the dynamic model. The controller gains used throughout the experimental evaluation and simulation are shown in Table II.

TABLE II: Position Controller Constants

Constant	Experimental Value	Simulation Value
K_x	18	1.5
K_y	10	0.07
K_θ	5	0.05
K_v	3	0.05
K_ω	1	0.05

2) Current Controller: The reference received by this block corresponds to the wheel torque, which is converted into a current reference using an identified motor torque constant K_m. Current regulation is achieved using a proportional-integral-derivative (PID) controller. The control input is computed from the error between the reference current and the estimated motor current. The corresponding control law is given by

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt}, \qquad (16)$$

where $e(t)$ denotes the current tracking error, and K_p, K_i, and K_d are the proportional, integral, and derivative gains, respectively [22]. The controller implemented on the STM32 microcontroller employs the gains $K_p = 7.5$, $K_i = 0.0001$, and $K_d = 0$.

D. Safety Controller

The proposed safety controller is formulated using a CBF and a CLF. The barrier and Lyapunov functions are defined as

$$B(x) = D - d_{\min} - t_f v, \qquad V(x) = (v - v_{\mathrm{obj}})^2, \qquad (17)$$

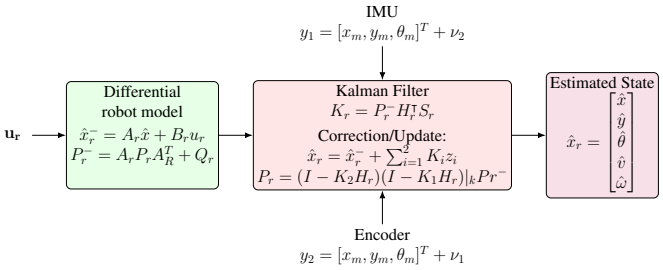


Fig. 3: Sensor fusion architecture.

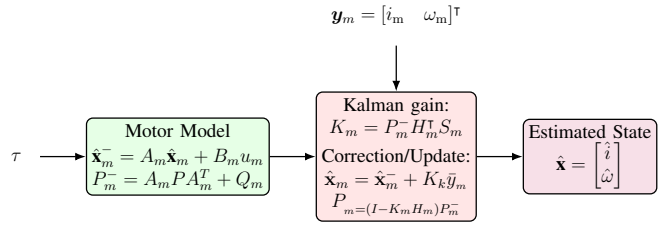


Fig. 4: Kalman Filter.

where v denotes the robot's linear velocity, D is the distance to the target, $d_{\min}$ is the minimum allowable safety distance, t_f represents the braking time, and v_{obj} is the desired linear velocity.

E. Kalman Filter

1) **Sensor fusion for system odometry:** In order to reduce measurement noise and compensate for model uncertainties, a Kalman filter was implemented in the Jetson AGX Orin that fuses information from the IMU, the odometry of the encoders, and the linearized model of the system to estimate the state of the robot Fig. 3. From the characterization of the sensors and the uncertainty of the model, the covariance matrices of the measurement noise were obtained $R_{\mathrm{IMU}} = \mathrm{diag}(10^{-2}, 10^{-2}, 10^{-2}, 5{\times}10^{-2}, 10^{-4})$, $R_{\mathrm{ENC}} = \mathrm{diag}(10^{-4}, 10^{-4}, 5 \times 10^{-3}, 10^{-3}, 2.5 \times 10^{-2})$, and the uncertainty matrix of the model $Q = \mathrm{diag}(10^{-2}, 10^{-2}, 10^{-2}, 10^{-2}, 10^{-2}, 5 \times 10^{-3}, 10^{-2})$.
2) **Current estimation using a Kalman filter**: On the NUCLEO-F446RE microcontroller, a second Kalman filter was implemented to estimate the motor current based on the motor model and the current sensor measurements Fig. 4. The covariance matrices, determined through system characterization, were $Q = \mathrm{diag}(5 \times 10^{-3}, 10^{-2})$ and $R = \mathrm{diag}(10^{-3}, 10^{-3})$

F. Experimental platform

Fig. 5 shows the platform used for the experimental validation of the proposed strategy. The system is composed of an NVIDIA Jetson AGX Orin, where high-level control and sensor fusion algorithms are implemented, and an STM32 NUCLEO-F446RE microcontroller responsible for acquiring current and encoder measurements, as well as for low-level motor control. Both devices communicate via ROS2 and Micro-ROS.

The platform also incorporates an Intel RealSense D435i camera, which provides images, depth data, and IMU measurements, an X-NUCLEO-IHM04A1 H-bridge for controlling the motors with incremental encoders, and two current sensors for system monitoring.

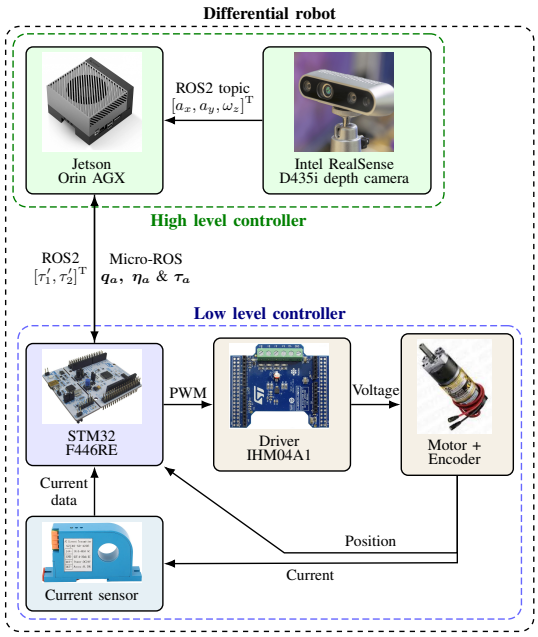


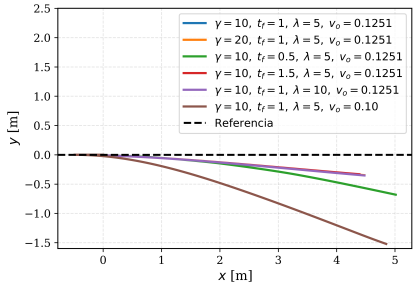
Fig. 5: Differential robot architecture.

IV. Simulation

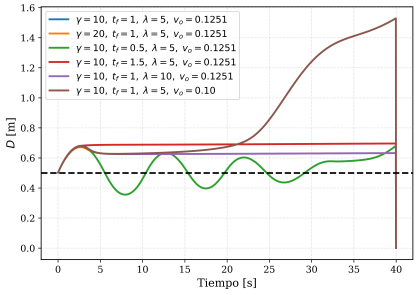
Fig. 6 presents the results obtained during the simulation process with parametric variation. These include the trajectory followed and the distance with respect to the target, the evaluation of the barrier function and Lyapunov conditions, and the relaxation variable. The results correspond to the tracking of a 5-meter trajectory, in which the robot takes the target position as a reference.

Fig. 6a shows the robot trajectories under different parametric variations along with the target trajectory. Fig. 6b displays the tracking distance, where the dotted line represents the minimum safety distance. The results demonstrate that the system maintains this constraint in most cases, except when the braking time is 0.5 s, a condition in which the robot repeatedly crosses the safety limit. This behavior is due to the fact that, with a shorter braking time, the barrier is less conservative, allowing a closer approach to the target and requiring greater control efforts when the barrier condition is activated.

To evaluate the effectiveness of the CBF, a target-tracking simulation was performed without enforcing the hard constraint imposed by the barrier function. The simulation setup assumes tracking the human target along a path of 1 m. The results shown in Figure 7 demonstrate that, in the absence of the safety filter, the robot collides with the human.

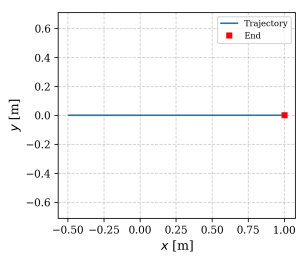


(a) Robot and target trajectories.

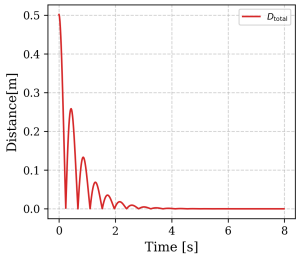


(b) Relative distance between the robot and the target.

Fig. 6: Simulation results of the proposed target-following controller.



(a) Robot trajectory.

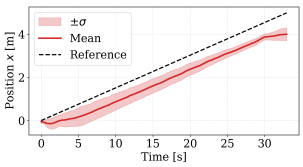


(b) Relative distance between the robot and the target.

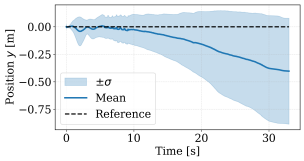
Fig. 7: Robot trajectory and relative distance between the robot and the target without the safety filter.

V. Experimental Results

This section presents the experimental results used to evaluate the proposed architecture. The experiment consisted of tracking a virtual target along a trajectory of 5 m on the x-axis, repeating ten times under the same operating conditions. The results include the mean and standard deviation of the robot's trajectory, the following distance, the applied torques, the barrier and Lyapunov conditions, and the relaxation variable.



(a) Trajectory along the x-axis



(b) Trajectory along the y-axis.

Fig. 8: Robot trajectory.

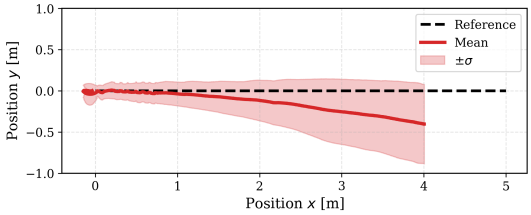


Fig. 9: Mean robot trajectory over ten experimental trials.

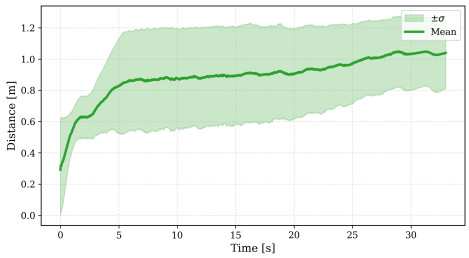


Fig. 10: Distance between the robot and the target during the tracking experiment.

Figures 8, 9 and 10 allow visualizing the path followed by the system during the experimental tests, as well as the distance relative to the reference throughout the entire trajectory. The results show that when the system is at the edge of the barrier, it attempts to move away from the reference in order to maintain a safe distance; once this occurs, the mobile robot begins to perform the assigned tracking task, preserving the distance and remaining above the defined threshold.

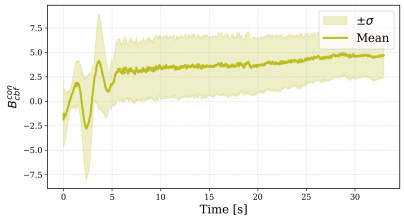
With regard to Fig. 11, the behavior of the Lyapunov and barrier conditions during the followed trajectory is evident. It can be observed that once the plant maintains a distance greater than the defined safety distance, the relaxation variable and the Lyapunov condition begin to converge toward zero, bringing the system's velocity state to a stable point in accordance with the imposed constraint.

TABLE III: Performance and safety metrics across ten experimental trials.

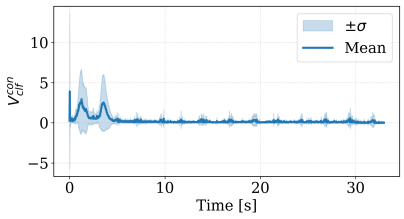
Metric	**Definition**	**Mean ± Std**
Violation magnitude	$\max(0, -B(t))$	0.357 ± 0.224
Completion time (s)	Elapsed time until completion criterion	28.97 ± 2.94
Maximum relaxation	$\max_t \delta(t)$	20.36 ± 21.64
Control effort	$\int_0^T (\tau_1^2 + \tau_2^2)\, dt$	0.228 ± 0.187

Table III summarizes the ten experimental trials. The mean barrier violation magnitude was 0.357 ± 0.224, potentially reflecting initial communication and optimization transients. The mean completion time was 28.97 ± 2.94 s, indicating relatively consistent task durations, whereas the maximum relaxation exhibited greater variability, 20.36 ± 21.64, possibly due to differences in initial conditions. The control effort was

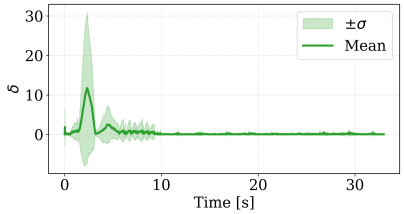
quantified by integrating the squared wheel torques over the task duration.



(a) Control Barrier Function condition.



(b) Control Lyapunov Function condition.



(c) Relaxation variable.

Fig. 11: Results of the stability constraint during the tracking experiment.

VI. Conclusions

This work developed a control strategy for safe navigation using a differential-wheeled mobile robot. The proposed strategy combines CLFs and CBFs within an optimization-based framework that solves a QP problem at each sampling instant. This approach provides safety guarantees for tracking tasks in which maintaining a safe distance from the virtual target is a critical requirement. The proposed controller was evaluated through both simulation and experimental tests, demonstrating robust behavior in the presence of parameter uncertainties. Although performance limitations associated with the estimation of the robot's orientation were identified, the proposed control strategy exhibited effective and consistent behavior throughout the evaluation. Future work includes implementing a real human tracking by incorporating a vision node.

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